IRRIGATION, INPUTS, CREDIT, INSURANCE AND SUSTAINABLE AGRICULTURE

IRRIGATION ACCOUNTING LADDER → PMKSY DISTINCTION BOARD → MICRO-IRRIGATION REBOUND → WATER-ENERGY-CROP NEXUS → WATERSHED RAIL → GROUNDWATER GOVERNANCE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 IRRIGATION ACCOUNTING LADDER

IRRIGATION ACCOUNTING LADDER · CREATED POTENTIAL · UTILISED POTENTIAL · RELIABLE FIELD DELIVERY · WATER PRODUCTIVITY · FARM OUTCOME

CREATED POTENTIAL

UTILISED POTENTIAL → RELIABLE FIELD DELIVERY → WATER PRODUCTIVITY / FARM OUTCOME

CONCEPT / ACCOUNTING BASIS
- CREATED POTENTIAL
- UTILISED POTENTIAL

TRANSMISSION / RECONCILIATION
- RELIABLE FIELD DELIVERY
- WATER PRODUCTIVITY / FARM OUTCOME

DISTRIBUTIONAL / STABILITY LIMIT
- CREATED POTENTIAL
- UTILISED POTENTIAL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RELIABLE FIELD DELIVERY; WATER PRODUCTIVITY / FARM OUTCOME.

01

STAGE 01 PMKSY DISTINCTION BOARD

PMKSY DISTINCTION BOARD · HAR KHET KO PANI · PER DROP MORE CROP · APPLICATION EFFICIENCY · STATE PLAN · DECENTRALISED PROJECT CHOICE · SOURCE TO FIELD

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
HAR KHET KO PANI → access	STATE PLAN → decentralised project choice	HAR KHET KO PANI → access
PER DROP MORE CROP → application efficiency	CONVERGENCE → source to field	PER DROP MORE CROP → application efficiency

MEASURE / INSTITUTION
- HAR KHET KO PANI → access
- PER DROP MORE CROP → application efficiency

DECISIVE DISTINCTION
- STATE PLAN → decentralised project choice
- CONVERGENCE → source to field

VINTAGE / LEGAL LIMIT
- HAR KHET KO PANI → access
- PER DROP MORE CROP → application efficiency

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STATE PLAN → decentralised project choice; CONVERGENCE → source to field.

02

STAGE 02 MICRO-IRRIGATION REBOUND

MICRO-IRRIGATION REBOUND · LESS WATER PER UNIT · LOWER PRIVATE COST · MORE AREA · WATER-INTENSIVE CROP MAY PERSIST · BASIN SAVING NOT AUTOMATIC

LESS WATER PER UNIT

LOWER PRIVATE COST / MORE AREA → WATER-INTENSIVE CROP MAY PERSIST → BASIN SAVING NOT AUTOMATIC

CONCEPT / ACCOUNTING BASIS
- LESS WATER PER UNIT
- LOWER PRIVATE COST / MORE AREA

TRANSMISSION / RECONCILIATION
- WATER-INTENSIVE CROP MAY PERSIST
- BASIN SAVING NOT AUTOMATIC

DISTRIBUTIONAL / STABILITY LIMIT
- LESS WATER PER UNIT
- LOWER PRIVATE COST / MORE AREA

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER-INTENSIVE CROP MAY PERSIST; BASIN SAVING NOT AUTOMATIC.

03

STAGE 03 WATER-ENERGY-CROP NEXUS

WATER-ENERGY-CROP NEXUS · CHEAP POWER · PROCUREMENT INCENTIVE · GROUNDWATER ACCESS · JOINT EFFECT · EXTRACTION + CROP LOCK-IN

CONCEPT / ACCOUNTING BASIS
- CHEAP POWER
- PROCUREMENT INCENTIVE

TRANSMISSION / RECONCILIATION
- GROUNDWATER ACCESS
- JOINT EFFECT → extraction + crop lock-in

DISTRIBUTIONAL / STABILITY LIMIT
- CHEAP POWER
- PROCUREMENT INCENTIVE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GROUNDWATER ACCESS; JOINT EFFECT → extraction + crop lock-in.

04

STAGE 04 WATERSHED RAIL

WATERSHED RAIL · VEGETATION + CONTOUR TREATMENT · BUNDS + MOISTURE RETENTION · RECHARGE STRUCTURES · PRODUCTIVE USE + MAINTENANCE

RIDGE → vegetation + contour treatment → MID-SLOPE → bunds + moisture retention → DRAINAGE → check / recharge structures → VALLEY → productive use + maintenance

CONCEPT / ACCOUNTING BASIS
- RIDGE → vegetation + contour treatment
- MID-SLOPE → bunds + moisture retention

TRANSMISSION / RECONCILIATION
- DRAINAGE → check / recharge structures
- VALLEY → productive use + maintenance

DISTRIBUTIONAL / STABILITY LIMIT
- RIDGE → vegetation + contour treatment
- MID-SLOPE → bunds + moisture retention

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRAINAGE → check / recharge structures; VALLEY → productive use + maintenance.

05

STAGE 05 GROUNDWATER GOVERNANCE

GROUNDWATER GOVERNANCE · PRIVATE PUMP · SHARED AQUIFER · LOCAL WATER BUDGET · CROP + EXTRACTION RULES

PRIVATE PUMP

SHARED AQUIFER → LOCAL WATER BUDGET → CROP + EXTRACTION RULES

CONCEPT / ACCOUNTING BASIS
- PRIVATE PUMP
- SHARED AQUIFER

TRANSMISSION / RECONCILIATION
- LOCAL WATER BUDGET
- CROP + EXTRACTION RULES

DISTRIBUTIONAL / STABILITY LIMIT
- PRIVATE PUMP
- SHARED AQUIFER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SHARED AQUIFER; LOCAL WATER BUDGET; CROP + EXTRACTION RULES.

06

STAGE 06 CREDIT-INSURANCE FORK

CREDIT-INSURANCE FORK · PRE-HARVEST LIQUIDITY · COVERED-LOSS TRANSFER · FINANCE ARCHITECTURE · TRIGGER + ASSESSMENT + SETTLEMENT

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
KCC → pre-harvest liquidity	NABARD / BANK → finance architecture	KCC → pre-harvest liquidity
INSURANCE → covered-loss transfer	CLAIM → trigger + assessment + settlement	INSURANCE → covered-loss transfer

MEASURE / INSTITUTION
- KCC → pre-harvest liquidity
- INSURANCE → covered-loss transfer

DECISIVE DISTINCTION
- NABARD / BANK → finance architecture
- CLAIM → trigger + assessment + settlement

VINTAGE / LEGAL LIMIT
- KCC → pre-harvest liquidity
- INSURANCE → covered-loss transfer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NABARD / BANK → finance architecture; CLAIM → trigger + assessment + settlement.

07

STAGE 07 INSURANCE OUTCOME CHAIN

INSURANCE OUTCOME CHAIN · PREMIUM + NOTIFIED SCOPE · LOSS ASSESSMENT · ADMISSIBLE CLAIM

PREMIUM + NOTIFIED SCOPE → LOSS ASSESSMENT → ADMISSIBLE CLAIM → SETTLEMENT

CONCEPT / ACCOUNTING BASIS
- PREMIUM + NOTIFIED SCOPE
- LOSS ASSESSMENT

TRANSMISSION / RECONCILIATION
- ADMISSIBLE CLAIM → SETTLEMENT
- PREMIUM + NOTIFIED SCOPE

DISTRIBUTIONAL / STABILITY LIMIT
- LOSS ASSESSMENT
- ADMISSIBLE CLAIM → SETTLEMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PREMIUM + NOTIFIED SCOPE; LOSS ASSESSMENT; ADMISSIBLE CLAIM → SETTLEMENT.

08

STAGE 08 BASIS-RISK BOARD

BASIS-RISK BOARD · AREA YIELD · LOWER VERIFICATION COST · FARM LOSS · MAY DIVERGE · WEATHER INDEX · PROXY TRIGGER · DATA + TIMELINESS DECIDE TRUST

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
AREA YIELD → lower verification cost	WEATHER INDEX → proxy trigger	AREA YIELD → lower verification cost
FARM LOSS → may diverge	DATA + TIMELINESS decide trust	FARM LOSS → may diverge

MEASURE / INSTITUTION
- AREA YIELD → lower verification cost
- FARM LOSS → may diverge

DECISIVE DISTINCTION
- WEATHER INDEX → proxy trigger
- DATA + TIMELINESS decide trust

VINTAGE / LEGAL LIMIT
- AREA YIELD → lower verification cost
- FARM LOSS → may diverge

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WEATHER INDEX → proxy trigger; DATA + TIMELINESS decide trust.

09

STAGE 09 SUSTAINABLE-PRACTICE MAP

SUSTAINABLE-PRACTICE MAP · MINIMUM DISTURBANCE + RESIDUE + ROTATION · DRIP + FERTIGATION + TENSIOMETER · INTEGRATED FARMING · CERTIFICATION AND TRANSITION CONTEXT

CONSERVATION → minimum disturbance + residue + rotation

PRECISION → drip + fertigation + tensiometer → DIVERSIFICATION → integrated farming → ORGANIC → certification and transition context

CONCEPT / ACCOUNTING BASIS
- CONSERVATION → minimum disturbance + residue + rotation
- PRECISION → drip + fertigation + tensiometer

TRANSMISSION / RECONCILIATION
- DIVERSIFICATION → integrated farming
- ORGANIC → certification and transition context

DISTRIBUTIONAL / STABILITY LIMIT
- CONSERVATION → minimum disturbance + residue + rotation
- PRECISION → drip + fertigation + tensiometer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONSERVATION → minimum disturbance + residue + rotation.

10

STAGE 10 TENANT-ACCESS GATE

TENANT-ACCESS GATE · ACTUAL CULTIVATOR · LEASE EVIDENCE · INSURANCE ELIGIBILITY · COVERAGE OR EXCLUSION

ACTUAL CULTIVATOR → RECORD / LEASE EVIDENCE → CREDIT / INSURANCE ELIGIBILITY → COVERAGE OR EXCLUSION

CONCEPT / ACCOUNTING BASIS
- ACTUAL CULTIVATOR
- RECORD / LEASE EVIDENCE

TRANSMISSION / RECONCILIATION
- CREDIT / INSURANCE ELIGIBILITY
- COVERAGE OR EXCLUSION

DISTRIBUTIONAL / STABILITY LIMIT
- ACTUAL CULTIVATOR
- RECORD / LEASE EVIDENCE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECORD / LEASE EVIDENCE; CREDIT / INSURANCE ELIGIBILITY; COVERAGE OR EXCLUSION.

11

STAGE 11 RESILIENCE ANSWER SPINE

RESILIENCE ANSWER SPINE · SEPARATE POTENTIAL, USE AND EFFICIENCY · LINK WATER, ENERGY, CROP AND CREDIT · TRACE ENROLMENT TO CLAIM SETTLEMENT · CONCLUDE WITH AQUIFER + TENANT GOVERNANCE

SEPARATE potential, use and efficiency → LINK water, energy, crop and credit → TRACE enrolment to claim settlement → CONCLUDE with aquifer + tenant governance

CONCEPT / ACCOUNTING BASIS
- SEPARATE potential, use and efficiency
- LINK water, energy, crop and credit

TRANSMISSION / RECONCILIATION
- TRACE enrolment to claim settlement
- CONCLUDE with aquifer + tenant governance

DISTRIBUTIONAL / STABILITY LIMIT
- SEPARATE potential, use and efficiency.
- LINK water, energy, crop and credit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRACE enrolment to claim settlement; CONCLUDE with aquifer + tenant governance.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · WATER PRICING, ELECTRICITY PRICING, PROCUREMENT INCENTIVES AND CROP CHOICE · MICRO-IRRIGATION SAVES WATER AT PLOT LEVEL, BUT BASIN-LEVEL SAVINGS · FORMAL TENANTS MAY GAIN CREDIT AND INSURANCE ACCESS, YET · AREA-YIELD INSURANCE REDUCES INDIVIDUAL VERIFICATION COSTS BUT INTRODUCES BASIS · CLIMATE ADAPTATION REQUIRES STRESS-TOLERANT SEEDS, ADVISORIES, WATER MANAGEMENT · PUBLIC EXTENSION IS A KNOWLEDGE INFRASTRUCTURE CONNECTING RESEARCH WITH · CENTRAL AND STATE IRRIGATION DEPARTMENTS

OPTIONAL MODEL / MEASUREMENT DEPTH
- CAUTION: Water pricing, electricity pricing, procurement incentives and crop choice form one
- CAUTION: Micro-irrigation saves water at plot level, but basin-level savings require limits on
- CAUTION: Formal tenants may gain credit and insurance access, yet insecure records create
- CAUTION: Area-yield insurance reduces individual verification costs but introduces basis risk.

VINTAGE, BASKET OR LEGAL LIMIT
- CAUTION: Climate adaptation requires stress-tolerant seeds, advisories, water management,
- CAUTION: Public extension is a knowledge infrastructure connecting research with local decision
- Central and state irrigation departments: build and manage major, medium and minor
- NABARD, banks and co-operatives: finance agricultural investment and seasonal

COMPARATIVE POLICY NUANCE
- Agricultural Insurance Company and empanelled insurers: implement crop-insurance
- ICAR, KVKs and state extension systems: provide research, advisories and farm-level
- CAUTION: Drip irrigation can reduce water applied per crop, but total aquifer extraction may
- CAUTION: Free power lowers the pumping cost seen by an individual farmer while accelerating

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels